Worked example SL14: Xu Jie's promotion of 1534

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Purpose of this example

This document follows one case, SL14, through the whole procedure of the thesis: the source passage, the information units listed from it, the three representations of the same passage, the judgement of every unit in every representation, and the competency questions that concern it. SL14 was chosen as a first example because it is short (ten information units) but still raises three of the main modelling problems: a promotion that implies leaving a previous post, a post that comes with an assignment, and an assignment that CBDB records as a separate office.

All judgements below are those of the frozen evaluation (rules v1, frozen 28 October 2026; evaluation tag `eval-SL14-v1.1`). They are reported as they stand, including where I would now model differently. A recheck made after the evaluation is reported separately in the last section but one.

Ministry, office and title names follow Hucker, *A Dictionary of Official Titles in Imperial China* (1985). In the Turtle excerpts, comments have been translated from Chinese and some lines omitted; the full files are listed in the last section.

The source passage

Ming Shizong shilu (Veritable Records of the Jiajing reign), juan 161, entry for the day *jimao*:

> 己卯，陞湖廣黃州府同知徐階為浙江按察司僉事，提調學校。

The quotation follows the page image, which reads 陞; the platform's transcription, and the information units listed below, have the variant 升.

"On *jimao*, Xu Jie, Vice Prefect of Huangzhou Prefecture in Huguang, was promoted to Assistant Surveillance Commissioner of Zhejiang, supervising schools."

Date. The year and month are not in the entry itself but in the heading of the juan: the third month of Jiajing 13, whose first day was *dingmao*. *Jimao* is therefore the 13th day of that month, which corresponds to 5 May 1534 in the proleptic Gregorian calendar (25 April 1534 Julian). The conversion was checked against the Academia Sinica calendar conversion tool.

Reading. The sentence has no grammatical subject; promotions were ordered by the emperor, so the issuer is implicit. The comma before 提調學校 was added by the modern editor of the text. The passage says that Xu Jie was promoted *from* one post *to* another, but it does not say that he left the first one; that is implied.

Background. In the Ming system a surveillance commission (按察司) in each province divided its work into circuits, one of which, in almost every province, was the Education Intendant Circuit (提督學道), in Hucker's translation (Hucker, 1958, p. 54). The circuit was assigned to one of the vice commissioners or assistant commissioners; it was not a post of its own (*Mingshi*, juan 75). Contemporaries called the official in charge of it a "supervising education official" (提學官), a term the same juan of the *Veritable Records* uses a few days later, when officials "formerly serving as supervising education officials of Guangdong" (原任廣東提學官) are punished (juan 161, Jiajing 13/3 yiyou, 11 May 1534). Title and function are therefore separate here, as they are for a Grand Secretary's ministerial title: the post of Assistant Surveillance Commissioner gives the rank, and supervising schools is a function attached to it. The model records the two separately, which is the general rule of decision D27.

Information units

The passage was broken into ten units, listed and approved before any modelling of this case began; the list was tagged `units-SL14` at that point. Two units are implicit (marked ✱): they are not written in the text but are clearly implied by it.

Unit	Type	Text	Note
J161-U01	date	嘉靖十三年三月己卯	year and month from the heading of the juan
J161-U02 ✱	issuer	(not written)	the emperor; the sentence has no subject
J161-U03	action	升 (promoted)	
J161-U04	title	湖廣黃州府同知 (Vice Prefect of Huangzhou)	the former post
J161-U05	name_form	徐階	
J161-U06	title	浙江按察司僉事 (Assistant Surveillance Commissioner of Zhejiang)	the new post

Unit	Type	Text	Note
J161-U07 ✱	relation	升……為…… (promoted from … to …)	from the old post to the new one; leaving the old post is implied
J161-U08	title	提調學校 (supervising schools)	an assignment expressed as a verb phrase, not an office title
J161-U09	relation	浙江按察司僉事，提調學校	the assignment is held by virtue of the new post
J161-U10	source	世宗實錄 卷161	

The units marked as *core* in the evaluation are U04, U06, U07, U08 and U09: they carry the modelling problems this case was chosen for.

Representation A: the existing CBDB records

Xu Jie is CBDB person 68547. The records relevant to this passage are four postings in the table of office holding (POSTED_TO_OFFICE_DATA), with their places in POSTED_TO_ADDR_DATA. The records were first read in a later SQLite release (`cbdb_20261003`) and then compared with the official standalone release of 2 June 2026 (file `cbdb_20260613`), which is the reference version of this project; the records cited here are identical in both.

Posting	Office code and name	Years	Place	Other fields
432216	70974 同知 (Vice Prefect, generic)	– to 1534	5576 黃州府, which belongs to 5520 湖廣	no start year
472581	400010 府同知 (Vice Prefect of a prefecture)	none	5576 黃州府	a duplicate of 432216
424200	70944 提刑按察使司僉事 (Assistant Surveillance Commissioner)	1534–1536	5331 浙江	appointment type 0 (“unknown”); notes cite *Shilu* juan 161

Posting	Office code and name	Years	Place	Other fields
424201	70951 提學使, a Ming-period code in CBDB (CBDB's translation: "Commissioner of an Education Intendant Circuit")	1534–1539	none	notes: `[提學使](提調學校)(明實錄:世宗實錄;161卷)`

For postings 424200 and 424201, the start year is coded as Jiajing 13, but the month and day fields are empty, and the source field (c_source) points to the Academia Sinica name authority database (9602), not to the *Veritable Records*; the juan number appears only in the free-text notes.

Three features of these records matter for the evaluation:

- The office title has been normalised: the province is removed from 浙江按察司僉事, but Zhejiang is kept as the place of the posting, so it can still be queried.
- The assignment 提調學校 has been recorded as a separate office, 提學使, with its own years (1534–1539, longer than the post, 1534–1536). The original words survive only in the notes.
- CBDB has codes that would have fitted better and were not used: appointment type 52 陞 (promoted); office 70923 提調學校 (used 6 times elsewhere in the database) or 72462 提刑按察使司提學僉事 (used 7 times); office category 15 (concurrent assignment); and the *Veritable Records* as a source text (13593). This is why several losses below are judged as *omission* rather than as a *limit* of CBDB.

Representation B: the CRM event model

The event model follows the source: the promotion is an act, carried out by the emperor, with a date and a source. Following decision D23, one promotion is a parent activity whose parts are the joining of the new institution and the leaving of the old one. Following D24, the leaving is modelled even though the text only implies it, and marked as implicit. Following D16 and D25, the assignment is a separate activity node, located in the province and linked to the institution, and motivated by the promotion.

```
# Parent activity: the promotion (D23)
event:jj13-03-jimao-sheng a crm:E7_Activity ;
    rdfs:label "陞湖廣黃州府同知徐階為浙江按察司僉事，提調學校"@zh-Hant ;
    crm:P2_has_type type:action-sheng ;              # verb 升, "promoted"
    crm:P14_carried_out_by person:ming-shizong ;     # implicit: the text has
no subject
    crm:P11_had_participant person:68547 ;           # Xu Jie
```

```
    crm:P4_has_time-span ts:jj13-03-jimao ;          # 1534-05-05
    crm:P9_consists_of event:jj13-03-jimao-sheng-1 , event:jj13-03-jimao-
sheng-2 .

# Part 1: joining the Zhejiang Surveillance Commission (D3)
event:jj13-03-jimao-sheng-1 a crm:E85_Joining ;
    crm:P143_joined person:68547 ;
    crm:P144_joined_with group:zhejiang-anchasi .

pc:jj13-03-jimao-sheng-1-anchasi a crm:PC144_joined_with ;
    crm:P01_has_domain event:jj13-03-jimao-sheng-1 ;
    crm:P02_has_range  group:zhejiang-anchasi ;
    crm:P144.1_kind_of_member type:anchasi-qianshi .  # Assistant
Surveillance Commissioner

# Part 2: leaving Huangzhou Prefecture (D24, implicit)
event:jj13-03-jimao-sheng-2 a crm:E86_Leaving ;
    crm:P145_separated person:68547 ;
    crm:P146_separated_from group:huangzhou-fu .

# The assignment: supervising schools (D16, D25, D26)
event:chaiqian-68547-tidiao-xuexiao a crm:E7_Activity ;
    crm:P2_has_type type:tidiao-xuexiao ;
    crm:P14_carried_out_by person:68547 ;
    crm:P7_took_place_at place:zhejiang ;
    crm:P11_had_participant group:zhejiang-anchasi ;
    crm:P17_was_motivated_by event:jj13-03-jimao-sheng .

# Source (D7)
doc:shizong-shilu-161 crm:P70_documents event:jj13-03-jimao-sheng ,
    event:jj13-03-jimao-sheng-1 , event:jj13-03-jimao-sheng-2 ,
    event:chaiqian-68547-tidiao-xuexiao .

# The title string is kept verbatim as an appellation (D4)
name:guanxian-jj13-03-jimao-xu-jie-yuan a crm:E41_Appellation ;
    rdfs:label "湖廣黃州府同知徐階"@zh-Hant .
```

The model uses no extension property. Two things in it rest on my judgement rather than on the text: the emperor as the one who carried out the promotion, and the leaving event. A weakness is visible in Part 2: the leaving event says that Xu Jie left Huangzhou Prefecture, but not that he left it *as Vice Prefect*. In this model the former post survives only in the verbatim title string.

Representation C: the CRM state model

The state model uses the same standard but records only what a person was, not what happened (D15). It contains no events and, by definition, no time: CIDOC-CRM allows a time-span only on temporal entities, and neither a property nor a property class such as PC107 is one. Membership of an institution is expressed with P107 and its property class PC107, which can carry the kind of member. An assignment without an institution of its own is expressed with the extension property `mpv:holds_title` (D22).

```
# Membership of the Zhejiang Surveillance Commission, as Assistant
Commissioner
group:zhejiang-anchasi crm:P107_has_current_or_former_member person:68547 .
pc:zhejiang-anchasi-member-68547 a crm:PC107_has_current_or_former_member ;
    crm:P01_has_domain group:zhejiang-anchasi ;
    crm:P02_has_range  person:68547 ;
    crm:P107.1_kind_of_member type:anchasi-qianshi .

# Membership of Huangzhou Prefecture, as Vice Prefect
group:huangzhou-fu crm:P107_has_current_or_former_member person:68547 .
pc:huangzhou-fu-member-68547 a crm:PC107_has_current_or_former_member ;
    crm:P01_has_domain group:huangzhou-fu ;
    crm:P02_has_range  person:68547 ;
    crm:P107.1_kind_of_member type:fu-tongzhi .

# The assignment, through the extension property
person:68547 mpv:holds_title type:tidiao-xuexiao .
```

The same verbatim title strings as in the event model are kept as appellations, with the *Veritable Records* juan 161 as their source (P67).

The comparison with the event model is instructive in one place. Because PC107 has a "kind of member" property, the state model can say that Xu Jie was a member of Huangzhou Prefecture *as Vice Prefect*, which the event model's leaving event cannot. But the state model cannot say when, in what order, or by whose act.

Evaluation of all units

Each cell gives the judgement and, for units not fully kept, the cause. *Limit* means the representation cannot express the unit (for the models: CIDOC-CRM together with the modelling decisions in force); *omission* means it could but did not; *conv.* marks a limit caused by one of my own modelling conventions rather than by the standard. kept_ext = kept only through an extension property; kept_judg. = kept only through the modeller's judgement.

Unit	CBDB	CRM event	CRM state
U01 date	partial / omission	kept	lost / limit
U02 issuer ✱	lost / limit	kept_judg.	lost / limit (conv.)
U03 action 升	lost / omission	kept	lost / limit (conv.)
U04 former post	kept	partial / omission	kept
U05 name	kept	partial / omission	partial / omission
U06 new post	kept	kept	kept
U07 promoted from…to ✱	partial / limit	kept_judg.	lost / limit (conv.)
U08 supervising schools	**distorted** / omission	kept	kept_ext
U09 post with assignment	partial / omission	kept_judg.	lost / limit (conv.)
U10 source	notes_only / omission	kept	kept

Totals (10 units)	CBDB	CRM event	CRM state
kept / kept_ext / kept_judg.	3 / – / –	5 / 0 / 3	3 / 1 / 0
notes_only / partial / distorted	1 / 3 / 1	0 / 2 / 0	0 / 1 / 0
lost	2	0	5
cause limit (of which conventions)	2	0	5 (4)
cause omission	5	2	1

Key decisions explained

Every row of the evaluation tables carries its evidence (the CBDB record or the triples, checked by a script) and a note with the reason. Five decisions show how the rules were applied.

U08, supervising schools: CBDB *distorted*. Recording 提調學校 as the office 提學使 gives Xu Jie an extra, independent office that the source does not support: the treatise on offices lists no such post under the surveillance commission, only an Education Intendant Circuit (提督學道) in the charge of a vice or assistant commissioner (*Mingshi*, juan 75). As a formal title, 提學使 dates only from 1905, when the provincial education commissioners (學政) were renamed (Qingshi gao, juan 116, 「改學政為提學使」); neither the treatise on offices of the Mingshi nor Hucker (1985) has a Ming entry under this name. CBDB

nevertheless has two codes with this name: a Ming one (70951, used for eleven persons with start years between 1487 and 1556) and a Qing one (85417). A query on CBDB therefore returns a wrong statement, not merely an incomplete one. The cause is *omission*, because suitable codes existed (70923, 72462). In the first pass this unit was judged *partial*; I changed it to *distorted* in review, and the change is recorded with its reason.

U07, "promoted from … to …": three different results. In CBDB the two postings are unrelated rows; the years (the old post ends in 1534, the new one begins in 1534) allow one to infer that they met in the same year, but not their order or that one replaced the other. This is *partial*, and a *limit*: CBDB has no field linking one posting to the next. In the event model, the promotion is one activity with the joining and the leaving as its parts, so the relation is expressed; but the leaving was added by me on the basis of an implication, so the judgement is *kept_judgement*, not *kept*. In the state model both memberships are present, but without time not even their coexistence can be inferred: *lost*, a limit caused by my own convention of modelling no events (D15).

U04, the former post: the event model does worse than the state model. The event model records that Xu Jie left Huangzhou Prefecture, but the leaving event cannot carry the kind of member he was there (CRM has P144.1 for joining but no corresponding property for leaving). In the first pass this was judged a *limit*. In review I changed the cause to *omission*, because the model could have added a separate joining event of unknown date with "Vice Prefect" as the kind of member, and did not. The state model, through PC107 and P107.1, does record "Vice Prefect", so it scores *kept* here.

U09, a post that carries an assignment: kept only by judgement. The event model has no direct link between the post (a joining event) and the assignment (an activity node). The two are connected only indirectly: the assignment was motivated by the promotion, the promotion contains the joining, and the assignment involves the same institution. Reading "held by virtue of the post" from this path requires knowing my conventions (D25, D26), so the unit is *kept_judgement*. This is the open question recorded as T1: whether posts too should be modelled as activity nodes, so that the two could be linked directly.

U01 and U02 in the state model: two kinds of limit. Both are lost. The date is lost because CIDOC-CRM does not allow a time-span on a membership; this is a limit of the standard. The issuer is lost because the state model deliberately has no events; this is a limit of my convention (D15) and is counted as such.

Competency questions CQ8 and CQ9

Two of the ten competency questions concern this case. The questions and their gold answers were written from the sources and fixed (tag `cq-v1`) before any query was written. A result is judged *wrong* if it contains any statement that contradicts the sources,

even if its other points are right.

CQ8. In Jiajing 13 (1534), who supervised schools in Zhejiang, and what was his substantive post? Gold answer: Xu Jie, from *jimao* of the third month of Jiajing 13, as Assistant Surveillance Commissioner of Zhejiang; before that he was Vice Prefect of Huangzhou.

	Verdict	What the query returned
CBDB	wrong	Xu Jie can be found by joining his 提學使 record with a same-year posting in Zhejiang, which gives the Assistant Commissioner post. But the result lists 提學使 as an independent office held 1534–1539, which contradicts the source
CRM event	correct	Xu Jie, from 1534-05-05, supervising schools, in Zhejiang, within the Zhejiang Surveillance Commission; post Assistant Commissioner; left Huangzhou Prefecture. The post and the assignment are linked only indirectly (see U09)
CRM state	partly correct	Xu Jie holds "supervising schools" and two memberships (Zhejiang as Assistant Commissioner, Huangzhou as Vice Prefect). There is no place or date to filter on, so it cannot say which membership was his post at the time

CQ9. Before and after the third month of Jiajing 13, in which institution did Xu Jie serve? Gold answer: before, Huangzhou Prefecture (Vice Prefect); after, the Zhejiang Surveillance Commission (Assistant Commissioner).

	Verdict	What the query returned
CBDB	wrong	Vice Prefect (to 1534, Huangzhou); Assistant Commissioner 1534–1536 (Zhejiang); 提學使 1534–1539. The last row is again a statement the source contradicts. The order of the first two can only be guessed from years that meet in 1534
CRM event	correct	joined the Zhejiang Surveillance Commission (Assistant Commissioner), left Huangzhou Prefecture, on the same day; the leaving is marked as implicit

	Verdict	What the query returned
CRM state	partly correct	both institutions and both kinds of member, but no order

What this example shows

- **Most of CBDB's losses here are omissions, not limits.** Of the seven units CBDB does not fully keep, five could have been recorded with codes that already exist in the database. Only the issuer and the link between the two postings are beyond its structure.
- **CBDB's most serious problem is a distortion, not a loss.** Recording an assignment as an office makes queries return a statement the source contradicts, which is why CBDB answers both competency questions wrongly.
- **The event model keeps almost everything, but three of its successes depend on my judgement.** The implicit issuer, the implicit leaving, and the link between post and assignment all rest on conventions I introduced. They are reported as *kept_judgement* so that they are not counted as achievements of the standard itself.
- **The state model shows what is lost without a node for time.** It keeps the same institutions and kinds of member, and even does better on the former post, but it cannot order them, date them or say who made the change.

A recheck made after the evaluation

After the evaluation was frozen, I reread the CIDOC-CRM scope notes for the properties involved, in particular for the case of U04. In the evaluation, the fact that the leaving event cannot carry "as Vice Prefect" was attributed to CRM, which has P144.1 for joining but nothing equivalent for leaving (open issue T4). The examples in the scope notes of P144 and P146, however, model an office itself as a group: "the office of Leader of the USSR" and "the office of President of the United States" are instances of E74. Under that pattern, leaving the group "Vice Prefect of Huangzhou" already says in what capacity Xu Jie left, and no further property is needed. The gap therefore comes from my own convention (D3: the institution is the group, and the post is a kind of member), not from the standard. The other pattern has its own cost: every office becomes a separate group, which must then be linked to its institution.

The frozen judgements are not changed. The recheck is recorded in the repository (modelling decisions, section "Review of 2026-11-09") and will be reported in the thesis as part of the discussion of alternative patterns, together with a similar recheck of two other

open issues (T2, "in the capacity of", where P14.1 *in the role of* covers part of the meaning; and T3, "running an office", where P14 is more appropriate than P11 because it implies responsibility).

Next step for this case. Following your suggestion, I plan to encode this case as fully as possible within the CBDB schema, using the codes listed in Representation A. This will separate what the schema can do from what the existing records happen to contain.

Where to find everything in the repository

Repository: github.com/KikaYang/ming-prosopography-crm. It is a snapshot of the work at this stage; the commit numbers and version tags mentioned in the documentation refer to the full development history, which is kept privately and is available on request. The files used in each evaluation are kept, unchanged, under `evaluated_versions/`.

What	File
Information units (rows J161-U01 to U10)	`data/evaluation/info_units.csv`
CBDB extracts for the three main persons	`data/deep_cases/cbdb_postings_3persons.csv`
Event model, with the diagram	`docs/modelling/SL14_event.ttl`, `docs/modelling/diagrams/SL14_event.png`
State model	`docs/modelling/SL14_state.ttl`
Shared vocabulary (persons, institutions, types)	`docs/modelling/vocab.ttl`
Modelling decisions D1–D26 and open issues T1–T4, with the 2026-11-09 recheck	`docs/modelling/modelling_decisions_en.md`
Evaluation tables	`data/evaluation/cbdb_vs_units_SL14.csv`, `event_vs_units_SL14.csv`, `state_vs_units_SL14.csv`
Evaluation rules v1 and the record of every evaluation version	`data/evaluation/README_en.md`
Script that rebuilds the SL14 tables from the first pass and the recorded changes	`scripts/eval/sl14.py`

What	File
Competency questions, queries, raw results and verdicts	`data/evaluation/competency_questions.csv`, `cq_queries/CQ8*`, `cq_queries/CQ9*`, `cq_raw/`, `cq_results.csv`

In the development history, the tag `units-SL14` marks the state of the unit list before modelling, and `eval-SL14-v1` and `eval-SL14-v1.1` mark the two evaluation runs (the second was needed because a block of the state model had been left out of a commit; it is described in the evaluation notes).